Informe - Exercici 3: Compressio GIF

Assignatura: Tractament d'imatges
Data de generacio: 2026-02-18

Aquest informe resumeix la conversio de les imatges PNG a GIF, l'analisi de paleta de colors, el calcul de bytes per pixel i l'efecte visual d'anul-lar la component vermella (R=0) en les imatges 3, 7 i 8.

Respostes concretes A-F

a) Conversio a GIF
- image1.png -> image1.gif (fet)
- image2.png -> image2.gif (fet)
- image3.png -> image3.gif (fet)
- image4.png -> image4.gif (fet)
- image5.png -> image5.gif (fet)
- image6.png -> image6.gif (fet)
- image7.png -> image7.gif (fet)
- image8.png -> image8.gif (fet)

b) Paleta de colors del GIF (resum concret)
- image1.gif: 256 colors; mes frequent (99, 99, 99) (4093 vegades)
- image2.gif: 256 colors; mes frequent (34, 24, 23) (3625 vegades)
- image3.gif: 256 colors; mes frequent (255, 255, 255) (6993 vegades)
- image4.gif: 256 colors; mes frequent (99, 99, 99) (4262 vegades)
- image5.gif: 256 colors; mes frequent (255, 255, 255) (49555 vegades)
- image6.gif: 256 colors; mes frequent (225, 214, 227) (3588 vegades)
- image7.gif: 256 colors; mes frequent (255, 255, 255) (312141 vegades)
- image8.gif: 244 colors; mes frequent (254, 146, 1) (49566 vegades)
- Detall complet de la paleta al bloc de detall de l'informe.

c) Bytes per pixel al GIF
- image1.gif: 1.000 B/px
- image2.gif: 1.000 B/px
- image3.gif: 1.000 B/px
- image4.gif: 1.000 B/px
- image5.gif: 1.000 B/px
- image6.gif: 1.000 B/px
- image7.gif: 1.000 B/px
- image8.gif: 1.000 B/px

d) Colors al PNG original (llistat acotat)
- image1.png: 55117 colors; mes frequent (99, 99, 99) (3467 vegades)
- image2.png: 63558 colors; mes frequent (99, 99, 99) (2968 vegades)
- image3.png: 45558 colors; mes frequent (255, 255, 255) (6792 vegades)
- image4.png: 25567 colors; mes frequent (99, 99, 99) (3225 vegades)
- image5.png: 24470 colors; mes frequent (255, 255, 255) (49555 vegades)
- image6.png: 72079 colors; mes frequent (65, 59, 46) (1240 vegades)
- image7.png: 258 colors; mes frequent (255, 255, 255) (312141 vegades)
- image8.png: 2377 colors; mes frequent (254, 146, 1) (49566 vegades)
- Llistat acotat: Top 8 i Bottom 8 colors per imatge.

e) Bytes per pixel al PNG
- image1.png: 3.000 B/px
- image2.png: 3.000 B/px
- image3.png: 3.000 B/px
- image4.png: 3.000 B/px

- image5.png: 3.000 B/px
- image6.png: 3.000 B/px
- image7.png: 3.000 B/px
- image8.png: 3.000 B/px

f) Comparacio i diferencia visual (concreta)
- GIF: paleta indexada fins a 256 colors; en aquest conjunt surt 1.000 B/px.
- PNG RGB: mes colors i 3.000 B/px.
- Visualment, el GIF pot perdre suavitat en gradients/textures per quantitzacio.

Resum numeric (apartats c i e)

Fitxer	Colors GIF	B/px GIF	Colors PNG	B/px PNG
image1.gif	256	1.000	55117	3.000
image2.gif	256	1.000	63558	3.000
image3.gif	256	1.000	45558	3.000
image4.gif	256	1.000	25567	3.000
image5.gif	256	1.000	24470	3.000
image6.gif	256	1.000	72079	3.000
image7.gif	256	1.000	258	3.000
image8.gif	244	1.000	2377	3.000

Comparacio (apartat f)

El GIF fa servir una paleta indexada de fins a 256 colors i acostuma a 1 byte/pixel. El PNG RGB manté molta mes variacio cromatica i fa servir 3 bytes/pixel sense compressio. Visualment, en GIF poden apareixer perdues en gradients i textures fines per la quantitzacio de color.

Detall de image1.png

GIF: image1.gif | Colors: 256 | B/px: 1.000
PNG: image1.png | Colors: 55117 | B/px: 3.000
Top colors GIF (RGB - aparicions):
  (99, 99, 99) - 4093
  (224, 188, 41) - 3649
  (48, 43, 33) - 3348
  (96, 97, 73) - 2885
  (27, 24, 22) - 2837
  (197, 161, 30) - 2800
  (103, 106, 87) - 2777
  (67, 65, 52) - 2667
Bottom colors GIF (RGB - aparicions):
  (26, 27, 25) - 501
  (149, 120, 78) - 472
  (36, 33, 31) - 387
  (109, 135, 118) - 304
  (186, 142, 78) - 262
  (51, 45, 39) - 261
  (170, 136, 82) - 238
  (102, 102, 99) - 233
Top colors PNG (RGB - aparicions):
  (99, 99, 99) - 3467
  (0, 0, 0) - 768
  (28, 27, 24) - 719
  (30, 28, 26) - 694
  (26, 24, 22) - 685
  (24, 23, 20) - 655

(27, 25, 23) - 651
(20, 18, 19) - 618
Bottom colors PNG (RGB - aparicions):
(255, 255, 186) - 1
(255, 255, 192) - 1
(255, 255, 213) - 1
(255, 255, 218) - 1
(255, 255, 223) - 1
(255, 255, 229) - 1
(255, 255, 232) - 1
(255, 255, 245) - 1

Detall de image2.png

GIF: image2.gif | Colors: 256 | B/px: 1.000
PNG: image2.png | Colors: 63558 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(34, 24, 23) - 3625
(127, 107, 88) - 3485
(99, 99, 99) - 3255
(120, 99, 81) - 3171
(100, 84, 68) - 3150
(39, 31, 28) - 3145
(92, 71, 55) - 2992
(73, 79, 47) - 2929
Bottom colors GIF (RGB - aparicions):
(48, 49, 39) - 218
(40, 25, 27) - 177
(126, 97, 59) - 172
(136, 111, 65) - 133
(68, 79, 59) - 132
(75, 87, 66) - 84
(57, 91, 70) - 33
(56, 79, 62) - 25
Top colors PNG (RGB - aparicions):
(99, 99, 99) - 2968
(34, 24, 22) - 874
(35, 25, 24) - 851
(0, 0, 0) - 768
(37, 27, 25) - 733
(33, 23, 21) - 705
(44, 37, 34) - 704
(46, 39, 36) - 701
Bottom colors PNG (RGB - aparicions):
(255, 255, 114) - 1
(255, 255, 121) - 1
(255, 255, 122) - 1
(255, 255, 131) - 1
(255, 255, 133) - 1
(255, 255, 154) - 1
(255, 255, 165) - 1
(255, 255, 178) - 1

Detall de image3.png

GIF: image3.gif | Colors: 256 | B/px: 1.000

PNG: image3.png | Colors: 45558 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(255, 255, 255) - 6993
(99, 99, 99) - 4088
(129, 143, 120) - 3430
(53, 52, 52) - 2931
(120, 137, 150) - 2754
(138, 150, 127) - 2676
(85, 90, 104) - 2617
(225, 232, 211) - 2563
Bottom colors GIF (RGB - aparicions):
(246, 228, 184) - 551
(131, 145, 132) - 492
(251, 255, 247) - 460
(252, 255, 255) - 373
(52, 53, 58) - 371
(252, 252, 243) - 276
(131, 147, 134) - 93
(251, 250, 255) - 74
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 6792
(99, 99, 99) - 3243
(52, 52, 52) - 887
(51, 51, 51) - 886
(0, 0, 0) - 768
(54, 54, 54) - 738
(50, 50, 50) - 563
(55, 55, 55) - 511
Bottom colors PNG (RGB - aparicions):
(255, 254, 225) - 1
(255, 254, 227) - 1
(255, 254, 229) - 1
(255, 254, 230) - 1
(255, 254, 247) - 1
(255, 255, 199) - 1
(255, 255, 200) - 1
(255, 255, 204) - 1

Detall de image4.png

GIF: image4.gif | Colors: 256 | B/px: 1.000
PNG: image4.png | Colors: 25567 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(99, 99, 99) - 4262
(181, 183, 143) - 3145
(75, 77, 54) - 2870
(169, 168, 134) - 2760
(176, 175, 137) - 2744
(173, 176, 141) - 2612
(158, 154, 120) - 2519
(183, 182, 143) - 2507
Bottom colors GIF (RGB - aparicions):
(104, 104, 101) - 299
(251, 255, 235) - 263
(89, 70, 50) - 252

(99, 108, 108) - 178
(173, 170, 140) - 159
(172, 167, 140) - 123
(179, 184, 132) - 48
(177, 169, 117) - 16
Top colors PNG (RGB - aparicions):
(99, 99, 99) - 3225
(177, 176, 139) - 2267
(179, 177, 140) - 2110
(176, 175, 137) - 2086
(169, 168, 133) - 1955
(171, 169, 134) - 1869
(168, 166, 131) - 1831
(175, 173, 136) - 1683
Bottom colors PNG (RGB - aparicions):
(255, 233, 213) - 1
(255, 242, 212) - 1
(255, 245, 212) - 1
(255, 248, 221) - 1
(255, 252, 223) - 1
(255, 255, 211) - 1
(255, 255, 219) - 1
(255, 255, 220) - 1

Detall de image5.png

GIF: image5.gif | Colors: 256 | B/px: 1.000
PNG: image5.png | Colors: 24470 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(255, 255, 255) - 49555
(255, 255, 224) - 4340
(255, 255, 239) - 3677
(255, 255, 243) - 3629
(255, 255, 223) - 3493
(255, 255, 225) - 3461
(255, 255, 240) - 3300
(255, 255, 249) - 3257
Bottom colors GIF (RGB - aparicions):
(204, 181, 139) - 188
(239, 221, 168) - 167
(119, 102, 57) - 164
(104, 103, 73) - 161
(135, 110, 47) - 123
(254, 234, 190) - 122
(129, 108, 55) - 42
(249, 254, 240) - 3
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 49555
(255, 255, 224) - 4237
(255, 255, 239) - 3677
(255, 255, 243) - 3629
(255, 255, 223) - 3493
(255, 255, 225) - 3403
(255, 255, 240) - 3300
(255, 255, 249) - 3257

Bottom colors PNG (RGB - aparicions):
(255, 255, 176) - 1
(255, 255, 179) - 1
(255, 255, 185) - 1
(255, 255, 187) - 1
(255, 255, 192) - 1
(255, 255, 194) - 1
(255, 255, 198) - 1
(255, 255, 200) - 1

Detall de image6.png

GIF: image6.gif | Colors: 256 | B/px: 1.000
PNG: image6.png | Colors: 72079 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(225, 214, 227) - 3588
(194, 184, 172) - 3480
(105, 97, 73) - 3237
(219, 174, 1) - 3178
(98, 90, 66) - 3134
(94, 153, 163) - 3073
(101, 93, 70) - 2981
(164, 163, 170) - 2954
Bottom colors GIF (RGB - aparicions):
(64, 55, 46) - 314
(82, 122, 85) - 280
(68, 57, 41) - 244
(245, 203, 161) - 200
(226, 185, 102) - 199
(85, 104, 75) - 172
(212, 157, 91) - 162
(248, 226, 168) - 30
Top colors PNG (RGB - aparicions):
(65, 59, 46) - 1240
(255, 255, 255) - 1106
(66, 61, 47) - 972
(63, 58, 45) - 957
(65, 59, 48) - 934
(0, 0, 0) - 768
(66, 60, 50) - 736
(94, 92, 77) - 703
Bottom colors PNG (RGB - aparicions):
(255, 253, 242) - 1
(255, 254, 221) - 1
(255, 254, 223) - 1
(255, 254, 233) - 1
(255, 254, 237) - 1
(255, 254, 252) - 1
(255, 255, 227) - 1
(255, 255, 228) - 1

Detall de image7.png

GIF: image7.gif | Colors: 256 | B/px: 1.000
PNG: image7.png | Colors: 258 | B/px: 3.000
Top colors GIF (RGB - aparicions):

(255, 255, 255) - 312141
(0, 0, 0) - 19685
(254, 254, 254) - 2575
(1, 1, 1) - 1059
(2, 2, 2) - 833
(253, 253, 253) - 820
(3, 3, 3) - 709
(4, 4, 4) - 671
Bottom colors GIF (RGB - aparicions):
(161, 161, 161) - 144
(166, 166, 166) - 144
(132, 132, 132) - 143
(174, 174, 174) - 142
(123, 123, 123) - 141
(138, 138, 138) - 140
(147, 147, 147) - 139
(148, 148, 148) - 138
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 312141
(0, 0, 0) - 19685
(254, 254, 254) - 2575
(1, 1, 1) - 1059
(2, 2, 2) - 833
(253, 253, 253) - 820
(3, 3, 3) - 709
(4, 4, 4) - 671
Bottom colors PNG (RGB - aparicions):
(132, 132, 132) - 143
(174, 174, 174) - 142
(123, 123, 123) - 141
(147, 147, 147) - 139
(138, 138, 138) - 138
(148, 148, 148) - 138
(137, 138, 138) - 2
(139, 140, 140) - 1

Detall de image8.png

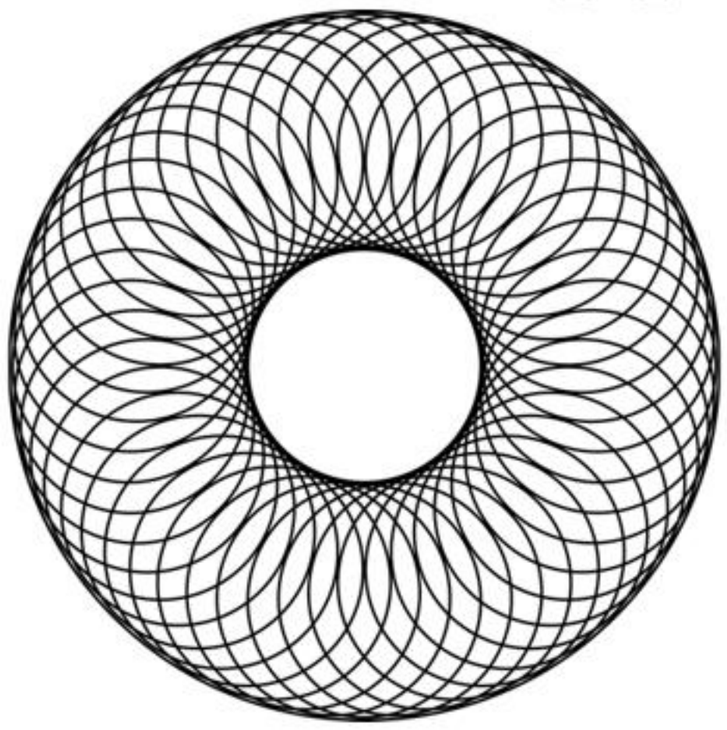
GIF: image8.gif | Colors: 244 | B/px: 1.000
PNG: image8.png | Colors: 2377 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(254, 146, 1) - 49566
(102, 190, 236) - 49542
(255, 229, 0) - 49515
(160, 92, 255) - 49023
(0, 113, 223) - 48402
(223, 16, 0) - 47718
(56, 167, 0) - 47282
(56, 167, 1) - 1116
Bottom colors GIF (RGB - aparicions):
(255, 228, 0) - 15
(102, 192, 237) - 14
(99, 192, 234) - 13
(102, 190, 238) - 12
(221, 11, 1) - 10

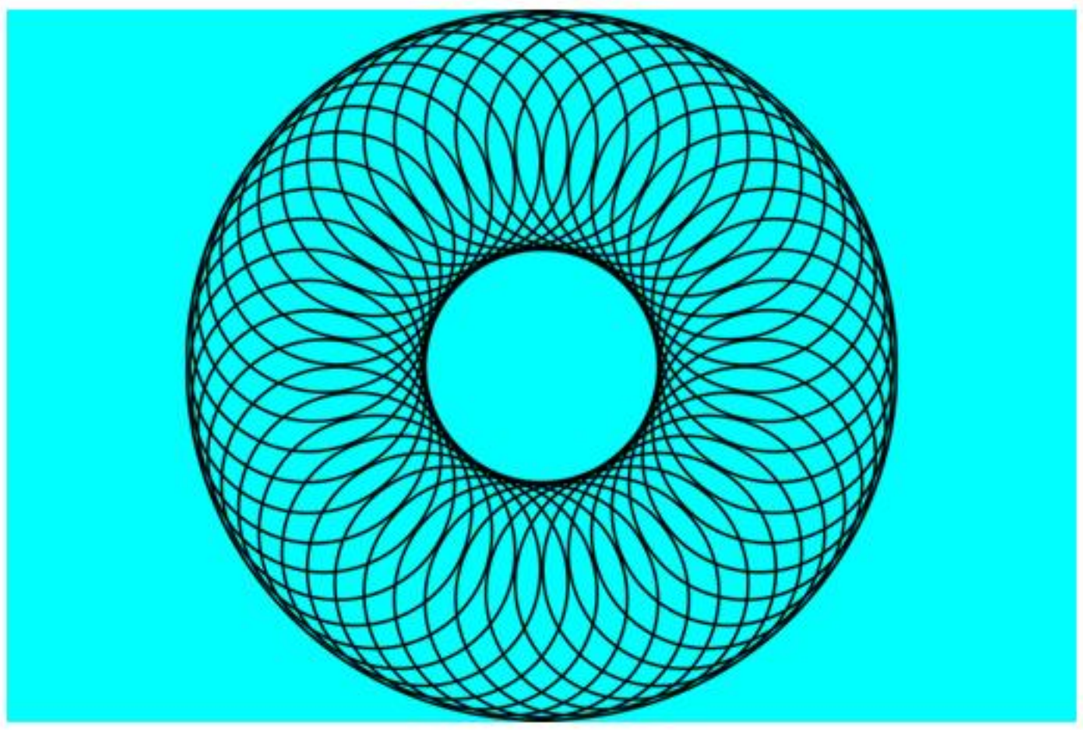
(104, 190, 237) - 5
(0, 110, 222) - 2
(253, 145, 0) - 2
Top colors PNG (RGB - aparicions):
(254, 146, 1) - 49566
(255, 229, 0) - 49515
(102, 190, 236) - 49505
(160, 92, 255) - 49023
(0, 113, 223) - 48402
(223, 16, 0) - 47718
(56, 167, 0) - 47277
(56, 167, 1) - 1116
Bottom colors PNG (RGB - aparicions):
(253, 139, 3) - 1
(253, 149, 3) - 1
(253, 154, 1) - 1
(254, 181, 2) - 1
(254, 186, 2) - 1
(254, 212, 1) - 1
(254, 227, 1) - 1
(255, 212, 1) - 1

Apartat g - Canvi de paleta (R=0)

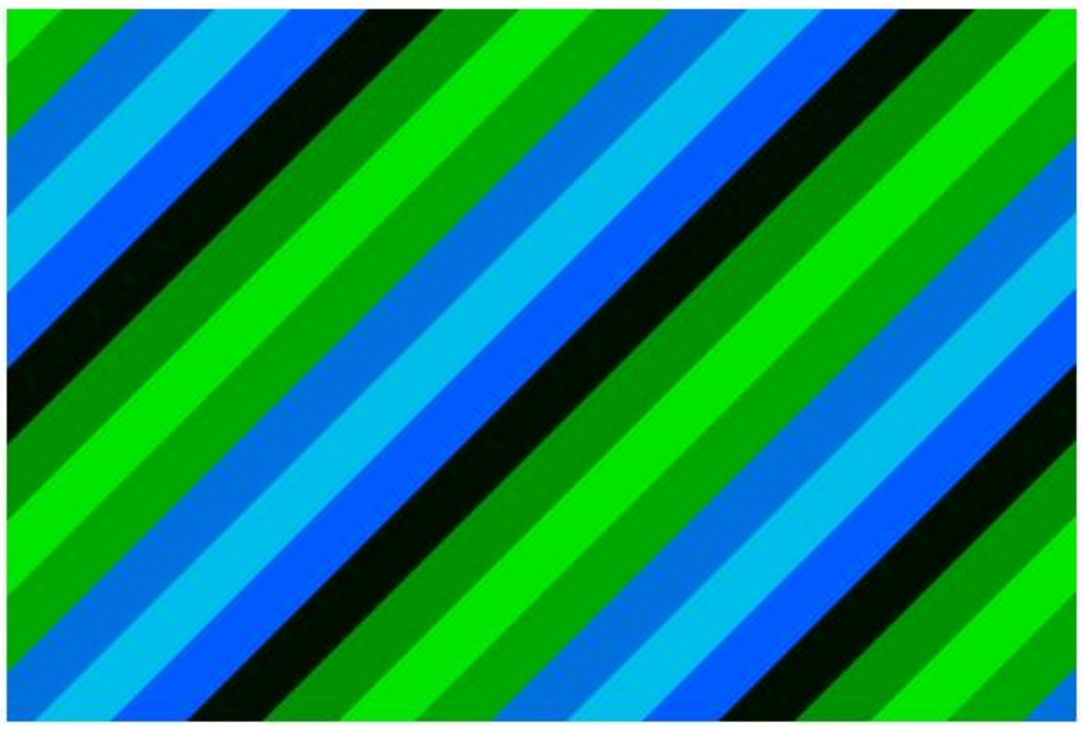
A continuacio es mostren les comparacions visuals entre el GIF original i el GIF modificat amb component vermella anul-lada (R=0) per a les imatges 3, 7 i 8.

Esquerra: image3.gif | Dreta: image3_no_red.gif





Esquerra: image7.gif | Dreta: image7_no_red.gif



Esquerra: image8.gif | Dreta: image8_no_red.gif

Annex - Codi utilitzat (A-F)

Aquesta seccio afegeix fragments de codi real utilitzat per obtenir les respostes dels apartats A-F, amb comentaris curts per entendre el flux.

1) Conversio PNG -> GIF (apartat a)

```
def ensure_gif_exists(png_path: Path) -> Path:
    gif_path = png_path.with_suffix('.gif')  # mateix nom, extensio GIF
    if not gif_path.exists():
        with Image.open(png_path) as png:
            png.save(gif_path, format='GIF')  # desa en color indexat
    return gif_path
```

2) Comptatge de colors del GIF (apartat b)

```
with Image.open(gif_path) as gif:
    if gif.mode != 'P':
        gif = gif.convert('P')  # assegura mode paleta
    palette = gif.getpalette()  # llista RGB de la paleta
    counts = Counter(gif.getdata())  # index de paleta -> aparicions
    used = sorted(counts.keys())
    colors = []
    for idx in used:
        rgb = tuple(palette[idx * 3: idx * 3 + 3])  # (R,G,B) del color
        colors.append((rgb, counts[idx]))
```

3) Bytes per pixel GIF/PNG (apartats c i e)

```
def gif_bits_per_pixel(num_colors: int) -> int:
    if num_colors <= 2: return 1
    if num_colors <= 4: return 2
    if num_colors <= 16: return 4
    return 8  # fins a 256 colors

gif_bpp = gif_bits_per_pixel(n_colors) / 8  # c)
png_bpp = 3.0  # e) PNG en RGB (3 canals x 1 byte)
```

4) Construccio de respostes A-F

```
def build_af_answers(summaries):
    answers = {}
    answers['a'] = [f"{png.name} -> {gif.name} (fet)" for png, gif, _, _ in summaries]
    answers['c'] = [f"{gif.name}: {gstats['bpp']:.3f} B/px" for _, gif, gstats, _ in summaries]
    answers['e'] = [f"{png.name}: {pstats['bpp']:.3f} B/px" for png, _, _, pstats in summaries]
    # b, d i f segueixen el mateix patro amb resumos concrets
    return answers
```